EACS Dataset: 4,407 real-world Bangladeshi admit card images with field-level ground truth for OCR and document understanding

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ABSTRACT

Educational institutions in South Asia issue large volumes of machine-printed examination admit cards, yet no openly available corpus exists for research on optical character recognition (OCR) or automated verification of such documents. This data paper presents the EACS (Exam Admit Card) Dataset: 4,407 admit card images collected between 2022 and 2025 from the NationalSchoolIntra enterprise resource planning platform operated by Al Quran Academy, Chattogram, Bangladesh. Each record is a standardised 800 × 460 pixel JPEG rendering of one card and carries nine textual fields (student name, father's name, mother's name, institutional student identifier, examination roll number, class, session, group and semester) together with the list of registered subjects and the scheduled examination date. A companion metadata file supplies a verified ground-truth transcription for every field of every card, so that recognition error can be measured at character, word and field level. The images were exported from the live production system rather than generated synthetically, and consequently retain the typographic, alignment and print-quality variation that synthetic corpora typically omit, a property that materially affects measured OCR accuracy. Student photographs, contact details and signatures were removed, and all personal names and identifiers were systematically randomised while preserving their structural and linguistic characteristics, allowing the corpus to be distributed openly. The dataset is released on Mendeley Data (version 1, doi: 10.17632/mvzdksjzvb.1). Intended applications include OCR benchmarking, document layout analysis, key-information extraction from administrative documents, and the development of tamper-detection mechanisms for academic credentials.

1. Introduction

Admissions and examination administration in large secondary schools is a high-volume, highly repetitive clerical process. At the institution from which the present corpus was drawn, examination admit cards are generated once per student per session (several thousand documents per examination cycle) and the resulting registers are transcribed into downstream information systems either by hand or through ad hoc optical character recognition (OCR) pipelines. Both routes are error-prone. Manual transcription introduces omissions and character confusions that are difficult to audit, while general-purpose OCR engines degrade noticeably on the typographic register of institutional forms: serif typefaces, inconsistent baseline alignment, mixed Latin and Arabic-derived numerals, and condensed tabular layouts (Smith, 2007; Reuel et al., 2019).

Reproducing that degradation in a laboratory is not straightforward. The benchmarks most often used to compare OCR systems are built from born-digital documents or from natural-scene photographs (Jaderberg et al., 2014), both of which differ substantially from a laser- or inkjet-printed administrative form emitted by an enterprise resource planning (ERP) system. Consequently, published accuracy figures transfer poorly to the documents that schools and universities actually process, and researchers who need realistic error estimates have no standard corpus against which to calibrate them.

This gap is wider for South Asian academic documents than the general literature suggests. The large public document-analysis corpora, for example PubLayNet (Zhong et al., 2019) and TableBank (Li et al., 2019), consist of scientific articles and are annotated at the level of layout regions and table structure, not at the level of individual administrative fields. Nothing in that collection is representative of a school examination register printed in English for a Bangladeshi institution, with its particular field order, class and group nomenclature, and subject coding scheme.

The EACS Dataset was assembled to close that gap. It provides (i) a representative collection of real, operationally generated admit card images; (ii) field-level ground-truth transcriptions aligned one-to-one with those images; and (iii) the anonymisation required for open redistribution. The present article describes the corpus in the format expected of a data descriptor (Elsevier, 2024), and states explicitly the limitations that users should account for when drawing conclusions from it.

2. Related work

2.1. Optical character recognition and its evaluation

Tesseract (Smith, 2007) remains the most widely deployed open-source OCR engine and is the default baseline in comparative evaluations; Reuel et al. (2019) benchmarked it against a range of contemporary alternatives

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and documented large variation in accuracy across engines, scripts and acquisition conditions. That variation is the reason benchmark corpora matter: an accuracy figure is meaningful only relative to the document population it was measured on. The present corpus is offered as one additional, explicitly described population (operational, low-complexity, South Asian academic forms) against which such comparisons can be run.

2.2. Sequence models and the role of representative training data

Modern recognisers are broadly sequence-to-sequence models, historically derived from bidirectional long short-term memory architectures (Graves and Schmidhuber, 2005), which replaced the hand-engineered character classifiers of earlier systems with learned representations. Learning those representations is data-hungry, and the field has therefore leaned heavily on synthetic corpora, most influentially SynthText (Ba et al., 2016), which renders arbitrary text into arbitrary backgrounds to achieve large scale. Synthetic generation solves the volume problem but introduces a domain gap: synthetic samples are smooth, uniformly contrasted and geometrically regular, whereas scanned or printed administrative documents contain speckle, toner dropout, baseline drift and inconsistent kerning. A model tuned predominantly on synthetic data can score well on synthetic held-out sets and noticeably worse on production documents. Corpora such as EACS are therefore needed as evaluation, and optionally fine-tuning, material.

2.3. Document layout analysis

Layout analysis, which partitions a page into semantic regions, has been served by several large annotated corpora. PubLayNet (Zhong et al., 2019) segments journal pages into text, title, list, table and figure regions; TableBank (Li et al., 2019) provides both detection and structure recognition for tables. Both derive from machine-readable academic articles. EACS complements them with a form-like document type: a single record per image, a fixed field inventory, and a stable reading order, which is the regime that key-information-extraction pipelines for administrative automation must handle.

2.4. Document tampering and credential verification

The same admit cards that are useful for OCR research are attractive targets for forgery, because a convincing credential can be produced by editing text within an otherwise genuine card. Statistical tests operating in the transformed or compressed domain have long been used to detect such manipulation (Fridrich and Kodovský, 2004; Goljan, Fridrich and Soukal, 2012). The forensic value of those tests depends on having authentic reference material drawn from the same issuing pipeline as the suspect document, which is precisely what an openly released corpus of unedited operational cards provides. Broader surveys of credential and phishing abuse (Almakhour et al., 2018) underline the practical motivation for automated verification of academic documents.

3. Data source and acquisition

Fig. 1 summarises how records moved from the institution's production system to the released corpus. Each stage is described in the subsections that follow. The two controls that either remove records or gate the release, the image-to-database check and the governance review, are drawn explicitly so that the composition of the final 4,407 records can be accounted for.

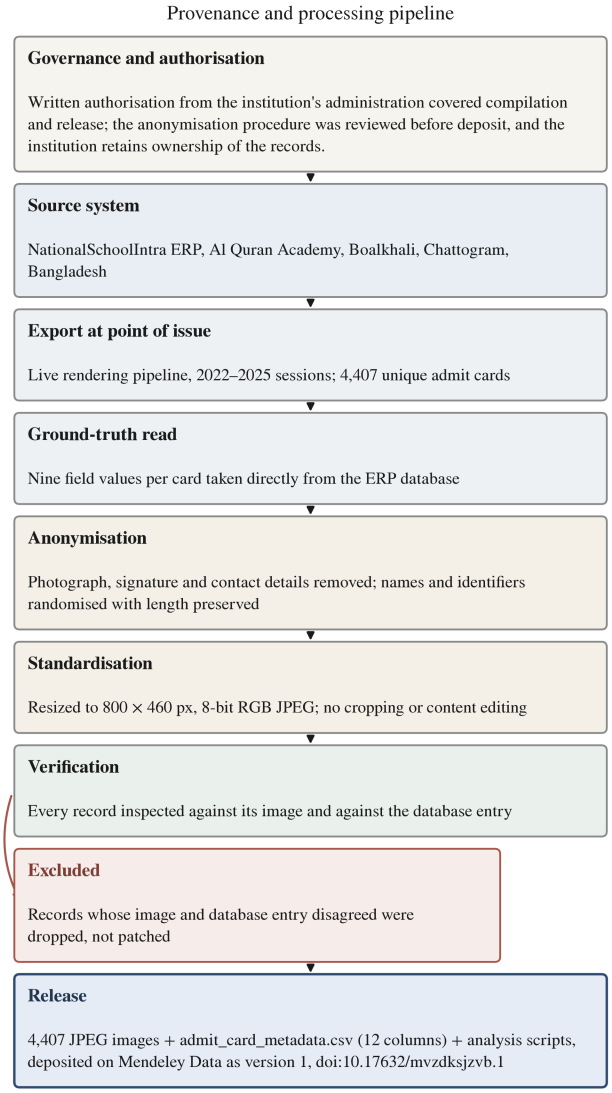


Fig. 1: Provenance and processing pipeline. Records were exported from the live NationalSchoolIntra ERP rendering pipeline at the moment of issue, read for ground truth from the same database, anonymised, standardised to a common canvas, and then checked against both image and database before release. Records whose image and database entry disagreed were excluded rather than patched.

3.1. Origin

All records originate from the NationalSchoolIntra ERP platform (NationalSchoolIntra, 2023), the production system used by Al Quran Academy, Boalkhali, Chattogram, Bangladesh, for student enrolment, examination scheduling and card generation. Cards were exported directly from the live rendering pipeline at the moment of issue, between the 2022 and 2025 academic sessions, yielding 4,407 unique records. Because the corpus is drawn from a running system rather than reconstructed for research, it inherits that system's template revisions, misaligned fields and print artefacts.

3.2. Authorisation

Compilation and release were carried out with the written authorisation of the institution's administration, which retains ownership of the underlying records and reviewed the anonymisation procedure described in Section 6 before release.

3.3. Standardisation

To keep the corpus homogeneous and storage-efficient, every exported record was resized to a fixed 800×460 pixel canvas and encoded as JPEG. Card templates share a common geometry, so this operation preserves the relative position and scale of all text regions; no cropping, rotation correction or content editing was applied. Field text was not redrawn or typeset by the authors; the images are the artefacts the system actually produced.

3.4. Analytical tooling

Summary statistics and all figures in this article were produced with the Python scientific stack (McKinney, 2010; Virtanen, Gommers, Oliphant et al., 2020) and Matplotlib (Hunter, 2007). The analysis scripts are distributed with the dataset so that every reported quantity can be regenerated from the released files.

4. Dataset description

Table 1 summarises the corpus. Table 2 gives the complete field inventory, and Figs. 2–6 characterise its composition.

4.1. Document layout

Fig. 2 shows a representative card and the field regions identified on it. The template divides into a branded header carrying the institution name and examination title, an identification block with the student's name, guardian names and institutional identifier, an examination block with roll number, class, session, group and semester, and a lower band listing the registered subjects with their numeric codes.

4.2. Field inventory and metadata schema

Nine textual fields are transcribed for every card: `student_name`, `father_name`, `mother_name`, `student_id`, `roll`, `class`, `session`, `group` and `semester`. These are the fields that vary per record and that an OCR pipeline must recover. Three further columns (the image filename, the scheduled examination date and the list of registered subjects) carry record-level rather than per-field information and support joining and date-aware analysis. Table 2 lists all twelve columns of the companion CSV.

The distinction matters for evaluation design. Because the template fixes field positions, a system may be scored in three ways: recognition accuracy on the character stream, word-level accuracy after segmentation, or field-level exact-match accuracy once the recognised string is assigned to a region and compared with the reference. The released ground truth supports all three because transcription is stored per field rather than as a single flattened page string (Section 5).

4.3. Composition of the corpus

Fig. 3 gives the number of cards per class and session. The corpus is not class-balanced: senior classes contribute more records than junior ones, and volume grows across sessions, reflecting cohort growth at the institution. Users who train models on this corpus should sample or reweight accordingly rather than assuming a uniform prior over classes.

Fig. 4 shows the distribution of academic group. Group assignment is categorical rather than continuous and is unevenly populated, so a model that predicts group from the printed text is solving a majority class problem.

The lower band of each card lists registered subjects, and the vocabulary of that band is the dominant source of lexical variety in the corpus. Fig. 5 ranks subjects by frequency. The head of the distribution is short: a small set of compulsory subjects accounts for most records, while the tail is long, containing optional and specialisation-specific subjects that appear in only a few cards. For OCR purposes this matters twice: frequent tokens dominate unweighted word error rate, so a recogniser can score well while failing on exactly the rare subjects that carry the most visual and lexical difficulty.

Finally, Fig. 6 shows the distribution of name lengths for students and their guardians. Names are the longest free-form strings on the card, and the spread of lengths indicates the range of segmentation difficulties a recogniser faces: short strings offer little context for language modelling, while long strings accumulate error over more characters. Because names in this corpus are synthetic substitutes of comparable length (see Section 6), the length distribution is faithful to the original even though the character content is not.

5. Ground-truth annotation and quality assurance

Ground truth was produced in two stages. First, the nine field values for each card were read directly from the ERP database, guaranteeing that the transcription matches what the issuing system believed it had printed. Second, every record was inspected against its image to confirm that the rendered card actually agrees with the database entry, and to catch records where the two had drifted apart, for example after a late correction to a roll number. Records failing this check were excluded rather than patched.

Table 1
Specifications of the EACS Dataset.

Item	Description
Name	EACS Dataset (Exam Admit Card Dataset)
Subject area	Document image analysis, optical character recognition and educational document verification
Type of data	Document images (JPEG) and tabular ground truth (CSV)
Data format	8-bit RGB JPEG, 800 × 460 pixels; UTF-8 CSV
How acquired	Exported from the production rendering pipeline of the NationalSchoolIntra ERP system at Al Quran Academy, Boalkhali, Chattogram, Bangladesh
Collection	4,407 admit cards issued across the 2022–2025 academic sessions
Ground truth	Field-level transcription for every card, verified against the issuing system
Anonymisation	Photographs, contact details and signatures removed; all names and identifiers randomised with structure and length preserved
Source	Al Quran Academy, Boalkhali, Chattogram, Bangladesh
Repository	Mendeley Data, version 1, doi: 10.17632/mvzdksjzvb.1

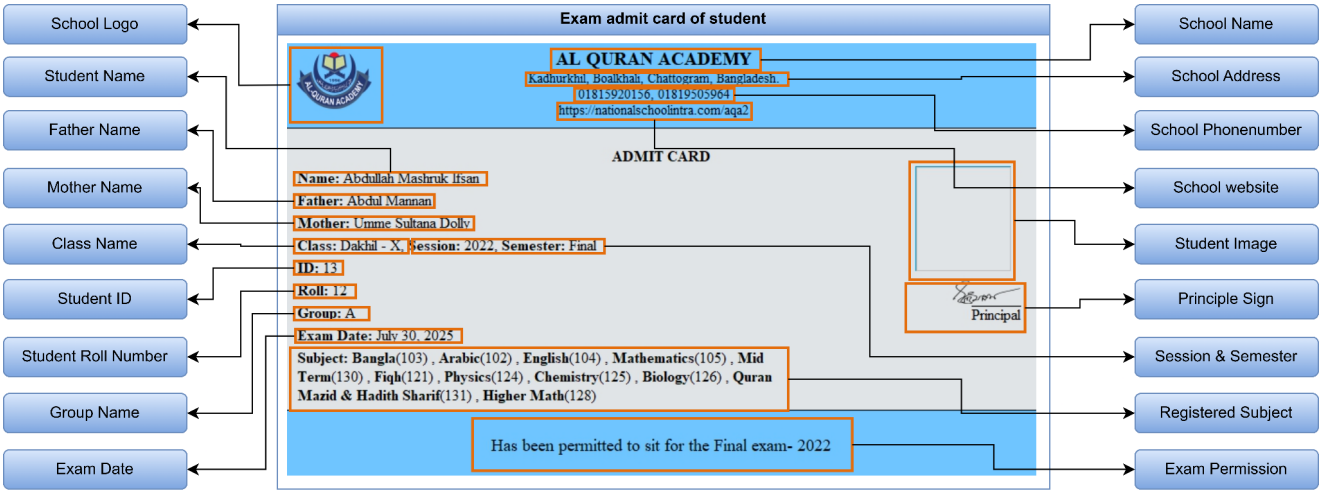


Fig. 2: Anatomy of an EACS admit card. The upper band carries institutional branding and the examination title; the central block contains the nine transcribed fields; the lower band lists registered subjects and their numeric codes. Field geometry is fixed by the card template, so the regions shown here are stable across the whole corpus.

Table 2
The twelve columns of the accompanying metadata file (`admit_card_metadata.csv`). One row per admit card image, keyed on `admit_card_image`. Nine columns hold the transcribed fields; the remaining three are record-level metadata.

Column	Type	Description	Example
`admit_card_image`	string	Filename of the card image; primary key.	`14-699-1-1-1-1.jpg`
`student_id`	integer	Institutional identifier (randomised).	`14`
`student_name`	string	Student name as printed (randomised).	Abid Anwar
`father_name`	string	Father's name as printed (randomised).	Mohammed Nurul Anwar
`mother_name`	string	Mother's name as printed (randomised).	Mosammat Sabekun Nahar Shafu
`roll`	integer	Examination roll number.	`13`
`class`	string	Grade level and examination stage.	Dakhil - X
`session`	integer	Session year of issue.	`2022`
`semester`	string	Examination stage.	Final
`group`	string	Academic stream or group (e.g. Science/Arts).	A
`exam_date`	date	Scheduled examination date, as printed.	30-Jul-25
`subjects`	string	Registered subjects with their codes, comma-separated.	Bangla (103), Arabic (102), English (104), Mathematics (105), …

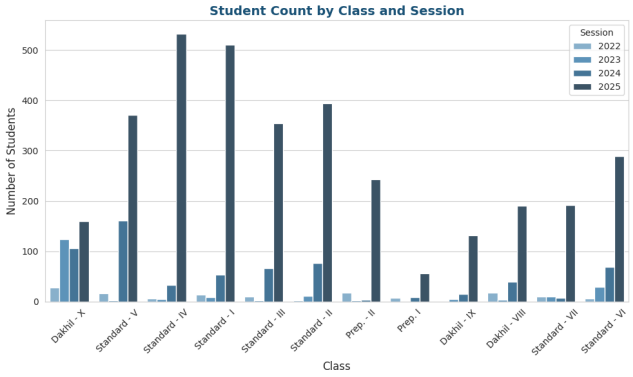


Fig. 3: Record count by academic class and session. Coverage is strongly non-uniform across grade levels and increases over the four sessions.

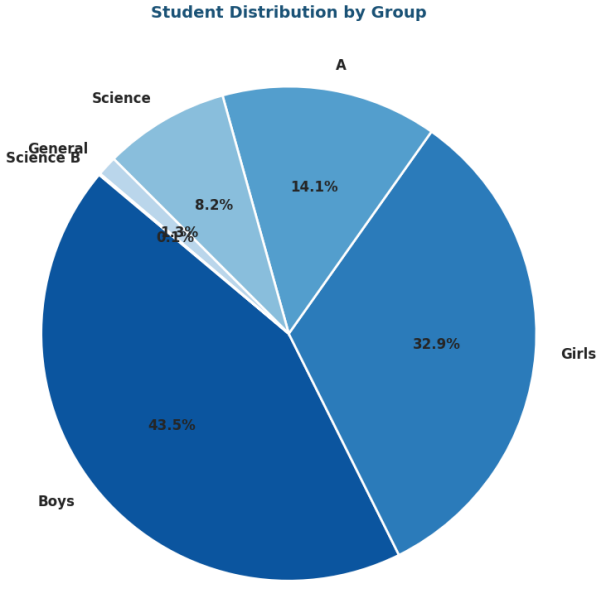


Fig. 4: Share of records by academic group. Group labels are short categorical strings and are heavily skewed.

Two limitations of this procedure should be stated plainly. The verification pass was performed by a single reviewer, so no inter-annotator agreement statistic is available; the ground truth is authoritative with respect to the issuing system but has not been independently replicated. And because transcription is recorded per field rather than per character span, no character-level bounding boxes are released. The fixed template makes field regions approximately recoverable, but users requiring exact localisation should derive and validate their own region definitions, and should measure field-level accuracy as well as recognition accuracy, since a correctly recognised string assigned to the wrong region is a failure mode that raw character error rate does not expose.

6. Ethics, privacy and consent

The corpus was compiled from records generated and retained by the institution for official academic purposes. Release was authorised by the institution's administration, which reviewed the procedure below before the data were deposited. The design follows the data-minimisation and purpose-limitation principles articulated in modern data protection regulation (European Union, 2016), applied here as a design reference for secondary research use of administrative data.

Table 3 lists the measures applied.

Table 3
Anonymisation and privacy measures applied before release.

Measure	Implementation
Field selection	Only the nine fields required for OCR and layout research are retained; no academic results, grades or attendance data are included.
Content removal	Student photographs, signatures, contact numbers and any other personal identifiers present on the source cards are removed.
Identifier randomisation	Names and institutional identifiers are replaced with synthetic substitutes matched to the original in token count, character class and script, so that length statistics and OCR difficulty are preserved.
Structure preservation	Field order, template geometry, class, group, session and subject codes are left unaltered, since these carry the research value of the corpus.
Release control	Deposited on Mendeley Data under a citable DOI and version number so that the released state is fixed and auditable.
Permitted use	Academic and non-commercial research. Re-identification of any individual, or use for commercial or administrative decision-making about identifiable persons, is out of scope.

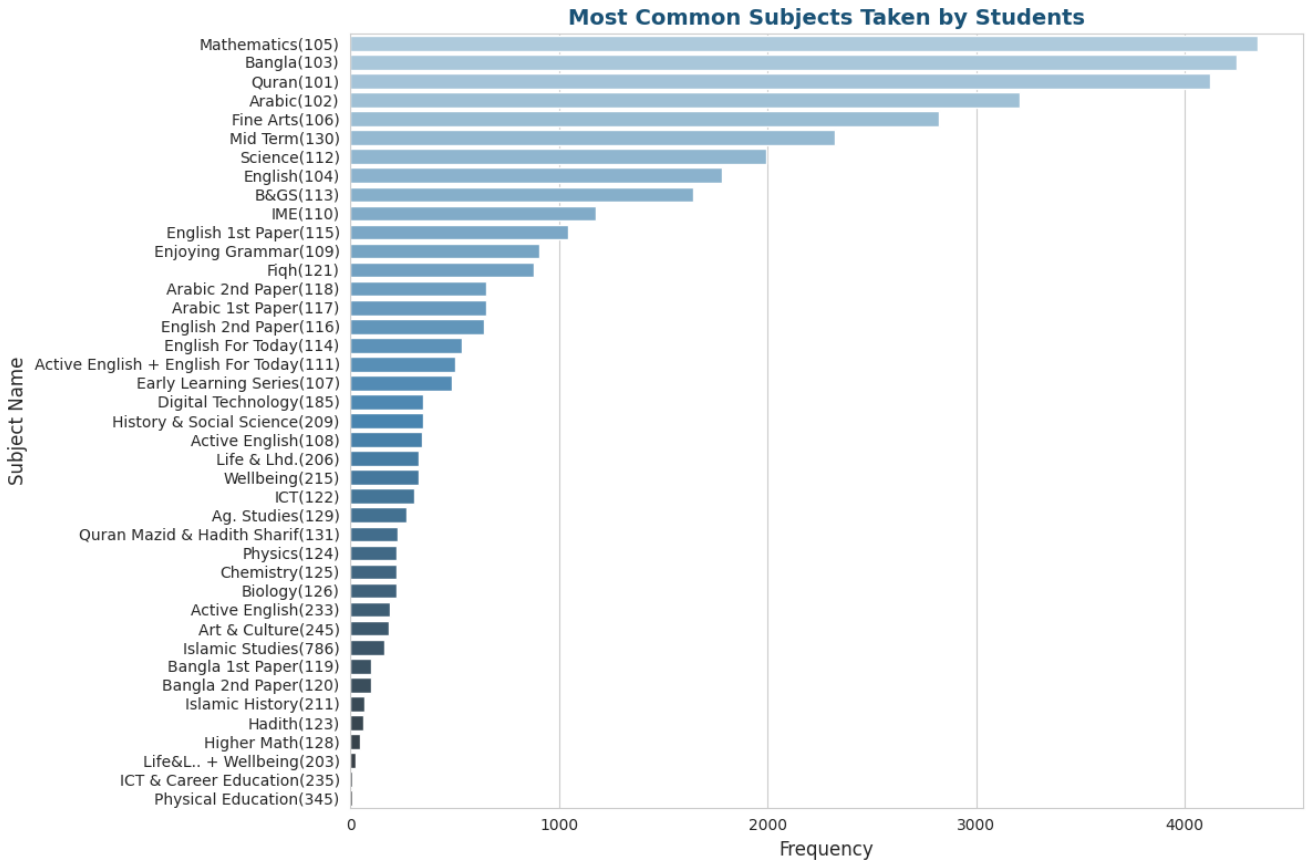


Fig. 5: Subject frequency across the corpus. The distribution has a short head of compulsory subjects and a long tail of optional subjects, so unweighted word error rate understates performance on the harder tail.

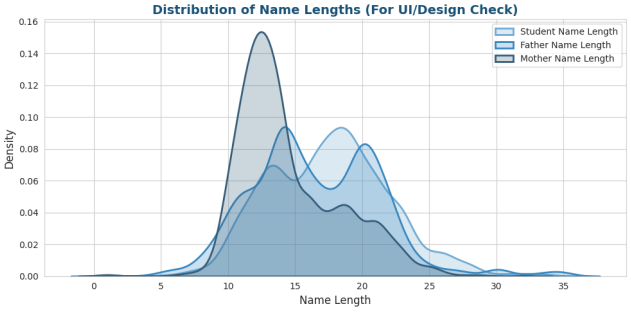


Fig. 6: Kernel density estimate of name lengths (characters) for students and their guardians, showing the range of free-form string lengths present in the corpus. Names are synthetic substitutes matched to the originals in length and character class (Section 6), so the length distribution is faithful to the original although the character content is not.

One residual risk should not be understated. Randomisation protects the identifiers themselves but not their context: the combination of class, session, group and subject list is distinctive enough that, in principle, a member of the original cohort could be narrowed down. The corpus should therefore be treated as pseudonymised rather than anonymised data, and users publishing derived results are advised not to report statistics at a combination of fields fine enough to isolate an individual record. No such attempt was made during preparation of this dataset.

7. Availability, versioning and reuse

The dataset is deposited on Mendeley Data as version 1 and is citable as Ashadzzaman (2026). Version numbers will be incremented for any correction to the released files; the DOI resolves to the most recent version while the version-specific URL remains resolvable in perpetuity. Users should cite the version they used.

The corpus is intended for research on OCR and document understanding: benchmarking recognition engines on production rather than synthetic documents; training and evaluating key-information-extraction models for fixed-schema administrative forms; studying layout analysis of form-like documents; and developing tamper-detection methods that require authentic reference material

from a real issuing pipeline. Distribution follows the findability and accessibility principles set out by Wilkinson, Dumontier, Aalbersberg et al. (2016), namely a persistent identifier, a public repository, and a documented schema.

8. Limitations

The following constraints bound what can legitimately be concluded from this corpus.

- **Single institution, single system.** All records originate from one school and one ERP configuration. Card templates elsewhere in the region will differ in field inventory, nomenclature and typography; results obtained here should not be assumed to transfer without re-evaluation.
- **English text only.** Bangla, Arabic and English subject names appear, but all personal names are Latin script. The corpus does not represent Bangla or Arabic orthography and cannot support work on those scripts.
- **Synthetic names.** Personal names were replaced. Length and character-class distributions are preserved, but the corpus cannot be used to study recognition of real Bangladeshi personal nomenclature, name matching, or name-based entity resolution.
- **No character-level ground truth.** Transcriptions are per field; exact character bounding boxes are not released (Section 5).
- **Fixed acquisition.** All images are 800×460 pixels in a single JPEG encoding. Performance at other resolutions, under recompression or on physical scans is not characterised by this corpus.
- **No photographs.** Faces were removed before release, so the corpus is unsuitable for any biometric task.
- **Class imbalance.** Coverage varies substantially across classes and sessions (Fig. 3), and unweighted accuracy estimates will be dominated by the largest groups.

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

The EACS Dataset is publicly available on Mendeley Data at https://data.mendeley.com/datasets/mvzdksjzvb/1, doi: 10.17632/mvzdksjzvb.1. Recommended citation: Ashadzzaman, Baizid MD (2026), "EACS Dataset: A Real-World Academic Admit Card Dataset from Bangladeshi Educational Institutions for OCR and Document Verification Research", Mendeley Data, V1, doi: 10.17632/mvzdksjzvb.1.

CRediT authorship contribution statement

Baizid MD Ashadzzaman: Conceptualization, Methodology, Software, Data curation, Validation, Investigation, Visualization, Writing: original draft, Writing: review and editing.

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